TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

Module Name:	Creative Engineering Design
Module Code:	SOC1020
Project Type:	Group Project (4-6 students)
Project Assigned Date:	02.03.2022
Team Registration Link:	Click Here
Team Registration Deadline:	19.03.2022 (midnight)
Deadline:	26.03.2022 (midnight)
Weighting:	20%
Instructor:	Dr. Sarvar Abdullaev
Email:	s.abdullaev@inha.uz
Office Hours:	Mon, 14:00 - 15:00
	Fri, 14:00 - 15:00
Office:	Room B411

Learning Outcomes

1. Analysing key principles of interaction for digital product and identifying human errors
2. Understanding basic concepts of User Experience (UX) research and design for digital products
3. Modelling simple and interactive User Interface (UI) for digital products
4. Understanding key steps in product design and development process
5. Modelling conceptual design of products and selecting conceptual models for further development
6. Prototyping and testing products according to their detailed specifications

Things to Submit:

1. **PRINTED and BOUND PRESENTATION SLIDES** as specified in this assignment
2. **PRINTED SUPPLEMENTARY MATERIALS** used or created during the meeting (e.g. customer interview transcripts, surveys, screenshots of competitor products, user flow diagrams, handwritten sketches or wireframes of your design concepts, FigJam files, Figma prototypes, or any other relevant research findings).
3. **ARCHIVE FILE** containing the electronic version of your presentation slides and all supplementary materials (e.g. customer interview transcripts, surveys, screenshots of competitor products, user flow diagrams, handwritten sketches or wireframes of your design concepts, FigJam files, Figma prototypes, or any other relevant research findings) **submitted to eClass.**

Design Project Overview

You are required to create the conceptual design of a digital product (e.g. mobile app, web app, or desktop app) that provides key UI models of different user experiences embedded in the functionality of your proposed project. Central to this project is a team-based approach to conceive and design a new digital product and present a prototype in the presentation session. The goal of this project is to learn principles and methods of digital product development to the point of conceptual design, to improve teamwork skills and to appreciate the inherent multidisciplinary nature of product development. You can pick an idea for your digital product from the list provided in Scenario section below. If your team has a different idea for a digital product which is not a mobile app, web app or desktop app, please validate your proposal with me by email or contact me during my office hours.

Scenario:

You have been employed by company X to develop the concept of a digital product they are going to develop and launch next year. You are asked to collect customer needs, research competitor products, compile product specifications, generate multiple concepts, select the most promising concept for the development of a prototype for user validation. Your key findings on completed user research and the final version of prototype should be presented to the top management of the company X as the final output of your project.

You can select the area of business of the company X from given list below, or come up with your own idea for a digital product.

- Hotel – customers can book rooms beforehand for certain period of time.
- Restaurant – customers can submit orders for food in menu and books tables.
- Publisher – subscribers can download books, authors can submit proposals to publish
- Taxi/Transfer – customers can book taxi beforehand indicating pick up and drop off locations.
- Online store – customers can buy goods online by adding items to their shopping cart and checkout at the end.
- Announcements board – users can post their announcements under different categories, moderators can modify, reorder or remove announcements.
- Personal blog – author can post blogs and subscribers can view and comment those blogs. As services, web site can advertise the skills and expertise of the author of the blog.
- Online surveys/tests – author can create surveys and tests, users can submit their answers and view results.

There should be a demonstrable market for the product. One good way to verify a market need is to identify existing products that attempt to meet the need. Your product need not be a variant of an existing product, but the market need addressed by your product should be clearly evident. The product does not need to have a tremendous economic potential, but should at least be an attractive opportunity for an established firm with related products and/or skills.

You should be confident of being able to prototype the product for a short period of time. Therefore you are urged to concentrate your efforts in specific needs of a small target market. Avoid addressing too many needs of big and complex markets. Moreover, you are not encouraged to choose product ideas which would assume a technological breakthrough (e.g. nanobots which can cure cancer) or unattainable market (e.g. people who live in Greenland and own penguins). Try to be realistic on both your market and technology for your proposed project. You should have access to more than five potential users of the product (more than 20 would be nice). You may also establish ties with an existing company that may be interested in your concept at the end of your project.

How to Register Myself to a Team?

Every student has to register himself as a member of some team in this Google Spreadsheet

Team Registration Link: Click Here

Teams cannot change their members after the deadline (1 week before official deadline) has passed. It is team leader's responsibility to ensure that their team members up-to-date in above spreadsheet.

Teams must contain at least 4 students and at most 6 students. Students from different sections cannot join the same team at the same time. This is due to the differences in their timetables which can be an issue while their project presentation.

What if I cannot find a team?

I will automatically assign those who could not find teams into new teams created by me. It is the responsibility of each student to check Google Spreadsheet to learn about his/her team after this assignment. Students who fail to participate in this team's activities **will get 0** for this assignment.

Steps to Complete the Project

You should organize group meetings every 2-3 days to assign tasks to members of the team and crosscheck on the progress of your project. Your meetings should be documented by filling out a **Meeting Minutes** enclosed as Appendix 1 (see section "How to Document Meetings"). Below are the key milestones of your project:

1. Register your team.
2. Formulate your problem scope and mission statement, assumptions and expected outcomes.
3. Plan your project's major tasks using Gantt chart
4. Research users. Conduct ethnographic interviews, analyse their interaction with a particular product using fundamental principles, find and classify human errors, create Personas (Appendix 2) and write Storyboards (Appendix 3) where your product will make an impact
5. Identify and interpret user/customer needs from obtained data
6. Formulate the problem statement for the project. Your problem statement should reflect your customer needs.
7. Define list of requirements for the product to be designed. Analyse competitor products, prioritize requirements that make the most impact using tools such as QFD matrix (Appendix 5). Compile the list of your product's target specifications.
8. Generate and select different User Experiences for your product. Construct multiple User Journey Maps (Appendix 4)/User Flows for suggested user experiences and then methodically select the most promising instances using Decision Matrix (Appendix 6). Use

concept generation techniques such as storyboarding with personas, SCAMPER, analogy, worst possible idea, brainstorming, mind mapping, morphology, biomimicry, etc.

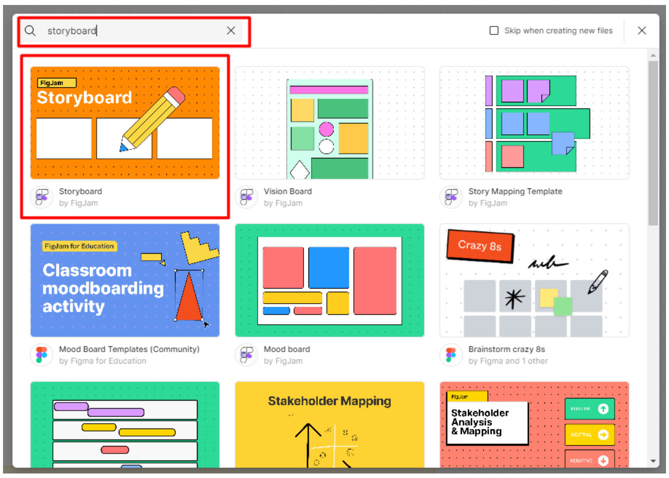
9. Generate and select low-fidelity UI screens for your chosen User Flow. Create multiple UI concepts for each step in the user flow and then methodically select the most promising UI screens using Decision Matrix (Appendix 6). Ensure that your UI complies with fundamental UX laws. Use concept generation techniques such as storyboarding, SCAMPER, analogy, worst possible idea, brainstorming, mind mapping, morphology, biomimicry, etc.
10. Design high-fidelity screens for your UI screens. Develop clickable prototypes for your selected user flow and UI sketches using Figma or any available technology.
11. Reflect on your design process. Find strong and weak points of your work and proposed concept.
12. Prepare presentation slides and practice your speech.
13. Submit electronic version of your presentation file, meeting minutes and all supplementary materials to eClass.
14. Print and bind your presentation file and meeting minutes along with relevant supplementary materials.

How to Document Meetings?

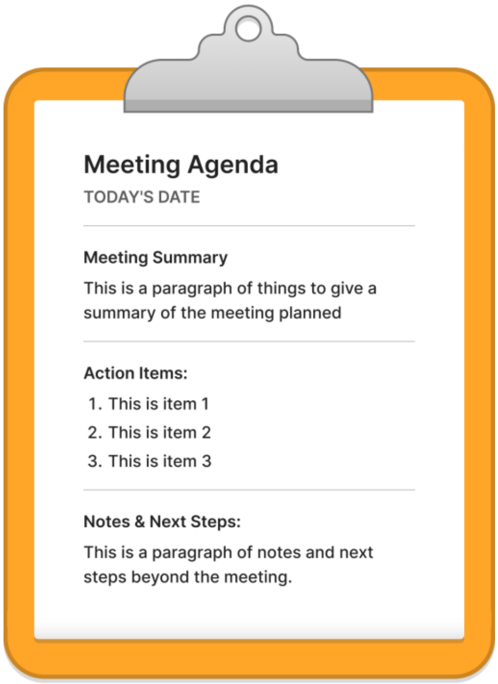
You can conduct meeting in two formats: online or offline. Each meeting has to be documented in an appropriate way as instructed below.

Online Meetings

Online meetings can be conducted in FigJam where you can brainstorm different parts of your project. FigJam has wide range of templates for creating Events Calendar, Personas, Storyboards, User Journey Map, Problem Statement, Prioritization Matrix, Product Specifications, Competitor Analysis, Flow Charts and UI Wireframes. In order to find the right template, click button, and search for a corresponding template:



Before each meeting, team leader is responsible to place this clipboard to the FigJam canvas and fill in the form. This meeting will be considered as an appropriately documented meeting.



Offline Meetings

Offline meetings should be documented by filling out a form attached in Appendix 1. Team leader has to prepare this meeting minute document before the meeting and ask each participant to sign to prove their presence. Also team leader should print out necessary templates provided in Appendices 2 – 6 for the upcoming meeting, so team members can discuss and fill out these templates in the process. All work produced during this offline meeting has to be stapled to meeting minutes document (as a cover page of this meeting) and added to your supplementary materials. You should submit them along with your presentation slides during your project's presentation.

How to Submit the Project?

You should archive your presentation slides and supplementary materials into a single archive file named as your team name (e.g. Lions.zip). This file should be submitted **only ONCE** by the team leader to eClass **before deadline**.

You should print out your presentation slides and bind them to a folder. For each meeting you have conducted with your team member, you should also print a meeting minutes document signed by attendees. Anything relevant used or created during your meeting should be enclosed to your meeting minutes document. All your printed and signed meeting minutes and their relevant attachments must be bound to a folder and submitted as supplementary materials to your presentation slides. **All your printed documents should be given to me before your presentation.**

How to Present the Project?

You should present your conceptual design of your digital product in a dedicated presentation session. Your team will be given **15 minutes** to present your work including the time for questions and answers. All of the team members must participate in presentation. If someone cannot participate in the presentation without excuse, he/she will **get 0 for this assignment.**

The structure of your presentation should be as shown below:

1. **Title page** contains team name, project title, team members' names and student IDs, section number.
2. **Mission statement page** contains the formulation of your goal, assumptions and expected benefits/objectives.
3. **Introduction page(s)** provides context to your product. This may involve brief explanation of the problem you are about to solve and relevant solutions that have solved similar problems.
4. **Customer needs page(s)** contains key highlights of your user research process and the list of prioritized customer needs.
5. **Product specifications page(s)** contains design metrics (functions) and their relationship to customer needs in QFD matrix. Final list of functions or their performance characteristics provided separately.
6. **Concepts page(s) for User Flows** contains key points of concept generation and selection process, and features important findings in User Flows.
7. **Concepts page(s) for UI screens** contains key points of concept generation and selection process, and features important findings in UI screens for your selected User Flow.
8. **Prototype page** demonstrates the clickable prototype of your product.
9. **Conclusion page** discusses about challenges and weak parts of the proposed concept.
10. **References page** provides links to prototype, and all other resources used in designing the UI.

Project Evaluation

Each member of the team will get the same mark given for the entire project. However, some member may get lower/higher mark too (see Disclaimer). Project will be evaluated according to following rubrics:

- Compliance with the product design process
- Reliability and thoroughness of customer needs research
- Feasibility of proposed product with current technology and the market
- Adequacy of design specifications and QFD
- Evolution of concepts from quantity to quality
- Originality of the proposed product
- Quality of the presentation
- Quality of the supplementary materials

The marks for this project will be distributed according to following scheme, but it does not mean that each component will be evaluated independently. Components of the project should be cohesive and serve the common goal of the project. Each assessment component of this project will be evaluated according to the rubrics mentioned above.

Assessment Component	Points
Introduction and problem scope	5
User research: interviews, personas, storyboards, human errors	10
User/Customer needs	10
Problem statement	5
Product specifications	10
Concept generation and selection: User Experiences	10
Concept generation and selection: UI screens (low-fidelity)	10
High-fidelity Prototype	10
Conclusion and self-evaluation	5
Oral presentation	15
Quality of supplementary materials	10
Total	**100**

Previous Year Projects:

- Promo videos on Youtube
- Presentation Files of Some Projects

Online Learning Resources

- Official Video Tutorials:
  - Figma Tutorials
  - FigJam Tutorials
- Popular Video Tutorials on Youtube:
  - Free Code Camp
  - Flux Academy
  - Code with Chris (Mobile App)

Penalties

Teams are subject to a penalty or disciplinary action for reasons of academic dishonesty and negligence. Academic dishonesty can be any attempt by a student to submit 1) work completed by another person outside the team without proper citation, 2) or give improper aid to another student outside the team in the completion of an assignment. Academic negligence can be disregarding the key requirements of the project including the timely submission of the project's deliverables. Following penalties will be exercises if any of the below cases happen:

- **Late submission.** If you missed the deadline, you must send your project deliverables over email to s.abdullaev@inha.uz within the next day after the deadline. You will be penalized by **20 points** for late submission. **Important notice:** you are not required to send your project deliverables by email if you have already submitted them to eClass before the deadline. If you submit your project deliverables to 2 places: eClass and email, you will get a penalty of **5 points** for confusing the assessment.
- **Making noise while others presenting.** If any member of your team is spotted for making noise while others presenting, the entire team will be penalized by **20 points**. If this repeats again, the team will be requested to leave the classroom and their mark will be 0.

- **Not informing about team change.** If you change a team or unable to find a team, you should inform me by email about this fact before team registration deadline. I will make changes in my teams list or assign you to a different team. If you fail to inform this fact or inform it after the deadline, you will be penalized by **10 points**.
- **Not participating in team's presentation.** If you fail to provide a mitigating excuse for your absence in your team's presentation, you will be considered as a non-contributor and **get 0 for this assignment**. If you have a mitigating excuse for your absence and it is planned ahead of time, you must notify me by email about this fact.

Any other academic misconducts including plagiarism, cheating, or breaching internal rules and regulations of IUT will be brought to the attention of Academic Committee for Penalty and Awards.

Disclaimer:

- I reserve the right to grade each member of the team separately if I have a suspicion on his/her contribution to the project. I will review meeting minutes submitted with the project, and if I find a member who has not contributed to the project, I will call this member for a short viva regarding his/her contribution. If member fails to show up, he/she will receive 0 for the project.
- I reserve the right to call the whole team for a short viva if I have suspicion about the quality of work submitted. If team fails to explain how web application has been implemented, each member of the team receive 0 for the project.

Appendix 1: Meeting Minutes

Attendance

<List of names here>

Meeting Location

Meeting Time

<Meeting start ~ end>

Agenda

<Agenda Item 1>

<Notes on discussion>

<Agenda Item 2>

<Notes>

Decisions Made

<Decision 1>

<Decision 2>

Post Meeting Action Items

<Action Item 1> <Assigned To> <Deadline>

<Action Item 2> <Assigned To> <Deadline>

Next Meeting Plan

<Location> <Date> <Time>

Appendix 2: Persona

<Name> | <Role>

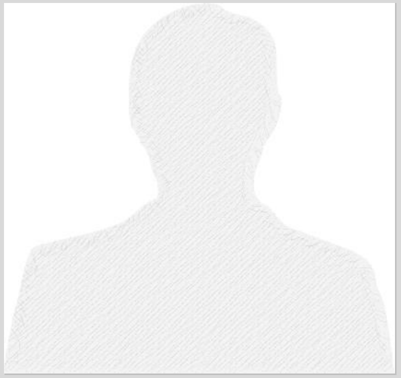
Profile: *100-200 word overview that describes the persona.*

Responsibilities: *Briefly describe the person's typical responsibilities.*

Pain Points: *List potential pain points that will affect this persona*

Key Drivers/Motivation: *What makes the persona go to work everyday? What makes the persona tick? What will the persona achieve?*

Validations: *Which input helps this person to make a decision? Is it product reviews? Analyses and reports? Or something else?*



Profile Attributes

Age:

Experience:

Personal details:

Common titles (AKA):

Verticals:

Company Size:

- Revenue range
- Employees
- Specifics subsets

Title: Page:

Scene No. Shot No.

Scene No. Shot No.

Scene No. Shot No.

Scene No. Shot No.

Scene No. Shot No.

Scene No. Shot No.

Appendix 3: Storyboard

Appendix 3: User Journey Map

User Journey Map: [Name of Persona]					
User Perspective *- Using the table below, map out the user's journey from the awareness stage to the advocacy stage. Each stage should be evaluated based on the user's goals, expectations, activities, and touchpoints with your organization. Then make a note of how satisfied users are at each stage of the experience.*					
	Awareness →	**Consideration →**	**Decision →**	**Delivery / Use →**	**Advocacy**
User Perspective					
User Goals					
User Expectations					
User Activities					
Touchpoints					
Overall User Experience	☐ Very satisfied ☐ Satisfied ☐ Neutral ☐ Dissatisfied ☐ Very dissatisfied	☐ Very satisfied ☐ Satisfied ☐ Neutral ☐ Dissatisfied ☐ Very dissatisfied	☐ Very satisfied ☐ Satisfied ☐ Neutral ☐ Dissatisfied ☐ Very dissatisfied	☐ Very satisfied ☐ Satisfied ☐ Neutral ☐ Dissatisfied ☐ Very dissatisfied	☐ Very satisfied ☐ Satisfied ☐ Neutral ☐ Dissatisfied ☐ Very dissatisfied

Appendix 5: QFD Matrix

		Desired direction of improvement (↑,0,↓)						
	1: low, 5: high	Functional Requirements (How's) →						
	Customer importance rating	Customer Requirements - (What's) ↓						Weighted Score
1								
2								
3								
4								
5								
6								
7								
8								
9								
		Technical importance score						
		Importance %						
		Priorities rank						

Appendix 6: Decision Matrix

Concepts Screening / Scoring

		A (reference)		B		C		D		E	
Selection Criteria	**Weight**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**
	Total Score										
	Rank										
	Continue?										